

Kaleem Nawaz Khan

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Areas of Interest: Networked Sensing, Autonomous Systems, Cooperative Perception

Research Focus	My research studies how vehicles and roadside infrastructure can collaboratively perceive complex environments beyond line-of-sight under strict latency, accuracy, and scalability constraints. A central theme of my work is the system-level co-design of perception, communication, and computation to enable reliable and safety-critical autonomous decision-making. My work has appeared in top mobile and sensing systems venues, including MobiSys and SenSys.	
Education	Rochester Institute of Technology (RIT) , Rochester, New York PhD Computing and Information Sciences Aug. 2022 – Present National University of Computer and Emerging Sciences , Peshawar, KPK Master of Science (Computer Science) Aug. 2015 – Dec. 2017	
Research Experience	Research & Development Center, General Motors , Warren, Michigan Research Intern (Mentor: Paolo Giusto) May. 2024 – Aug. 2024 - Developed solutions to enable smart off-loading of vehicular data to edge/cloud - Explored OMNET++ with Simu5G integration to investigate different network scenarios for vehicle-to-edge communication and data collection Networked Sensing Systems Lab, RIT , Rochester, New York Graduate Research Assistant (Advisor: Fawad Ahmad) Aug. 2022 – Present - I develop fast, accurate, and scalable infrastructure-assisted and vehicle-to-vehicle cooperative perception systems. - Additionally, I also explore collaborative, view-based 3D scene representations to enhance vehicles' perception beyond traditional approaches. National Center of Artificial Intelligence , Peshawar, KPK Research Team Lead (PI: M. Salman Khan) Apr. 2019 – Sep. 2020 - Led a diverse team in R&D to develop AI solutions for healthcare. - We utilized deep learning and related technologies for screening to assist cardiology and hematology practitioners.	
Research Projects	FRC: Radar–Camera Fusion for Intersection Awareness Radar, Camera, Homography, Multimodal Sensor Fusion Nov. 2025 – Present - Exploring solutions to map sparse radar data onto 2D images and fusing complementary information reliably. - Accurate Spatial Alignment and time synchronization to correctly match radar detections with camera frames. PMC: Collaborative Intersection Perception Using Multi-Vehicle Cameras SfM, 3D Reconstruction, Spatial Reasoning, Segmentation May. 2025 – Present - Creating and real-time updating SfM-based 3D models of the intersection using images from vehicle-mounted cameras. - Estimating dynamic objects in the scene based on object segmentation. - Performing spatial reasoning on the SfM-based 3D model to identify blind spots and determine the most relevant information for vehicles to exchange. ARC: Accurate, Real-time, and Scalable Multi-vehicle Cooperative Perception C++, LiDAR, PCL, CUDA, CarLA Jun. 2024 – Mar. 2025 - Enabled high-accuracy cooperative perception while preserving scalability, addressing a key gap in existing systems. - Added a refinement layer for vehicle point cloud alignment, reducing errors in existing methods. - Employed grid-based spatial reasoning to minimize alignment latency and enable selective data sharing among vehicles. Smart Vehicle-to-Edge Computation and Data Offloading OMNET++, Simu5G, MEC, ML May. 2024 – Aug. 2024	

- Enabled smart off-loading of vehicular data and apps to edge/cloud
- Used Simu5G to simulate complex 5G network scenarios involving moving vehicles, base stations, and edge servers.
- Generated simulated data on network latency, signal strength, and congestion under various conditions.

VRF: Vehicle Road-side Point Cloud Fusion

ROS, FAST-LIO, PCL, Open3D, Docker

Sep. 2022 – Mar. 2024

- Tackle vehicles' sensor occlusion by enabling real-time sharing and accurate fusion of raw 3D data from roadside LiDARs.
- Decoupled vehicle and roadside point cloud alignment by aligning both to a 3D map and then enabling a direct alignment.
- Evaluation on real-world testbeds demonstrates VRF can fuse vehicle roadside point clouds with 20ms end-to-end latency and 5 cm positioning accuracy, thereby multiplying the vehicle's reaction time by a factor of 5.

Publications

Conferences

- **Khan, K. N.**, & Ahmad, F. (2025). ARC: Accurate, Real-time, and Scalable Multi-vehicle Cooperative Perception. In *Proceedings of the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026)*.
- **Khan, K. N.**, Khalid, A., Turkar, Y., Dantu, K., & Ahmad, F. (2024, June). VRF: Vehicle Road-side Point Cloud Fusion. In *Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (MobiSys 2024)*.
- Hasib, R., **Khan, K. N.**, Yu, M., & Khan, M. S. (2021, April). Vision-based human posture classification and fall detection using convolutional neural networks. In *2021 International Conference on Artificial Intelligence (ICAI)* (pp. 74-79). IEEE.
- Qasim, S., **Khan, K.N.**, Yu, M. and Khan, M.S., 2021, April. Performance evaluation of background subtraction techniques for video frames. In *2021 International Conference on Artificial Intelligence (ICAI)* (pp. 102-107). IEEE.
- Ahmad, B., Khan, F.A., **Khan, K.N.** and Khan, M.S., 2021, December. Automatic classification of heart sounds using long short-term memory. In *2021 15th International Conference on Open Source Systems and Technologies (ICOSST)* (pp. 1-6). IEEE.
- Islam, M., **Khan, K.N.** and Khan, M.S., 2021, April. Evaluation of preprocessing techniques for U-Net based automated liver segmentation. In *2021 International Conference on Artificial Intelligence (ICAI)* (pp. 187-192). IEEE.

Journals

- **Khan, K.N.**, Khan, F.A., Abid, A., Olmez, T., Dokur, Z., Khandakar, A., Chowdhury, M.E. and Khan, M.S., 2021. Deep learning based classification of unsegmented phonocardiogram spectrograms leveraging transfer learning. *Physiological measurement*, 42(9), p.095003.
- **Khan, K.N.**, Ullah, N., Ali, S., Khan, M.S., Nauman, M. and Ghani, A., 2022. Op2Vec: An Opcode Embedding Technique and Dataset Design for End-to-End Detection of Android Malware. *Security and Communication Networks*, 2022(1), p.3710968.
- Salman Khan, M., Ullah, A., **Khan, K.N.**, Riaz, H., Yousafzai, Y.M., Rahman, T., Chowdhury, M.E. and Abul Kashem, S.B., 2022. Deep Learning Assisted Automated Assessment of Thalassaemia from Haemoglobin Electrophoresis Images. *Diagnostics*, 12(10), p.2405.
- Zeng, A., Wu, C., Lin, G., Xie, W., Hong, J., Huang, M., Zhuang, J., Bi, S., Pan, D., Ullah, N. and **Khan, K.N.**, 2023. Imagecas: A large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images. *Computerized Medical Imaging and Graphics*, 109, p.102287.

Selected Talks

VRF: Vehicle Road-side Point Cloud Fusion, Tokyo, Japan

ACM MobiSys 2024, International Conference on Mobile Systems, Applications, and Services
3 – 7 June. 2024